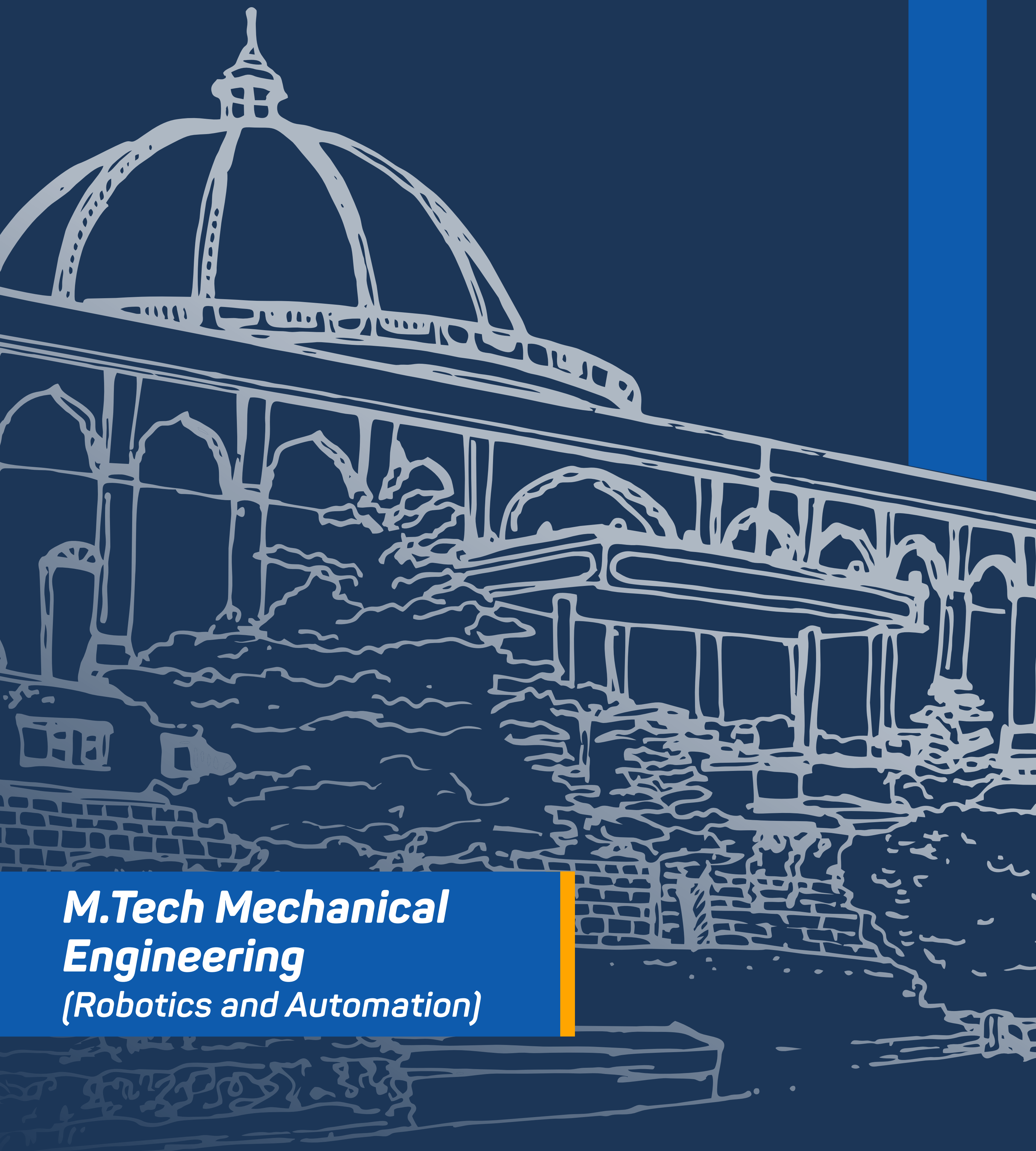
MIT-WPU

FACULTY OF ENGINEERING AND TECHNOLOGY



M.Tech Mechanical Engineering
(Robotics and Automation)

Programme Structure

mitwpu.edu.in

Division	Faculty of Engineering and Technology
School Name	School of Engineering & Technology
Department Name	Department of Mechanical Engineering
Programme Name	M.Tech. Mechanical Engineering (Robotics and Automation)

COURSE BASKET

Course Type	Description
Programme Core	Courses dealing with foundations, depth and breadth of the major in which student is admitted at MIT-WPU
Programme Electives	Open electives under the programme allow students to specialise in a particular area connected to their major.
University Core	Courses that reflect the core MIT-WPU values and the mission of Life Transformation of students.
University Electives	Multidisciplinary courses across the faculties at MIT-WPU and outside the programme core.

Semester I

Semester	Course Type	Course Name / Course Title	Total Credits
I	PC	Advanced Mathematics	3
I	PC	Fundamentals in Robotics	4
I	PC	Advanced Industrial Automation	4
I	PC	Control System in Robotics & Automation	4
I	PC	Research Methodology for Mechanical Engineers	4
I	PC	Robotics Lab	1
I	UC	Scientific Studies of Mind, Matter, Spirit and Consciousness	2
I	UC	Yoga - I	1
		Total	**23**

Semester II

Semester	Course Type	Course Name / Course Title	Total Credits
II	PC	Analysis and Synthesis of Mechanisms	4
II	PC	Artificial Intelligence and Machine Learning	3
II	PC	Internet of Things for Mechatronics	4
II	PE	Program Elective-I	4
II	PE	Program Elective-II	4
II	PR	Seminar	2
II	UC	Peace Building: Global Initiatives	2
		Total	**23**

Semester - III

Semester	Course Type	Course Name / Course Title	Total Credits
III	PC	Machine Vision System	4
III	PE	Program Elective-III	4
III	PE	Program Elective-IV	4
III	PR	Research Project - I	8
		Total	**20**

SEMESTER - IV

Semester	Course Type	Course Name / Course Title	Total Credits
IV	PR	Internship	4
IV	PR	Research Project - II	12
		Total	**16**

Program Electives

Semester	Course Type	Course Name / Course Title
II	PE - I	Robot Operating System
II	PE - I	Intelligent Robotic Systems
II	PE - I	Robot System Reliability and Safety
II	PE - I	Digital Manufacturing System
II	PE - II	Advances in Mobile Robots and AGV
II	PE - II	Medical Robots w
II	PE - II	Drone Technology
II	PE - II	Robotics in Healthcare & Agriculture
III	PE - III	Computer Aided Engineering
III	PE - III	Advanced Mechanical Vibration
III	PE - III	Soft Robotics
III	PE - III	Industrial Engineering & Management
III	PE– IV	Embedded System Design and RTOS
III	PE– IV	Industry 5.0 and Emerging Technologies
III	PE– IV	Modern Control Panel
III	PE– IV	Autotronics

*Modifications to the programmes and courses are contingent upon adherence to university guidelines and procedures. Any proposed changes must undergo a thorough review process, including consultation with relevant academic departments, approval from the appropriate administrative bodies, and compliance with accreditation standards.

Additionally, consideration will be given to feedback from students, faculty, and other stakeholders to ensure that modifications align with the overall educational objectives and mission of the university. The implementation of any approved changes will be communicated transparently to the university community, and appropriate measures will be taken to facilitate a smooth transition for all affected parties.